Derived Intersection Theory on Foliation

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In this paper, I'll discuss foliation from point of view of derived algebraic geometry. Derived algebraic geometry proves the ∞-equivalence of dg Lie algebroid and foliation. Now the quotient is defined by quotient of dg L_∞ groupoid generated from dg Lie albebroid, and it's calculated by dg L_∞ version of Chevalley-Eilenberg complex. I'm also wondering evaluation of the foliation using language of DT/GW invariants. First of all, the idea of foliation is classically defined differential geometrically on a smooth manifold:

Definition 0.1. Foliation

Foliation is a partition of a smooth manifold M into its submanifolds. One idea of generating foliation is to retrieve from distribution $D_x \subset TM$ for some $x \in M$, and this distribution classfies the points $x \in M$ to define foliaton. Distribution D_x is defined by its vector field $X, Y \in \Gamma(D_x)$ satisfying $[X, Y] \in \Gamma(D_x)$, and in this case D_x is called integrable. This D_x corresponds to a submanifold L, since there's its restriction $D_x|_L = T_x L$ by the unique maximal submanifold $L \subset M$ s.t. $x \in L$. Note in fact $D_x|_{L_y} = T_y L$ for all $y \in L$. Accordingly, the resulting set $\{L_y\}$ for $y \in M$ is the set of submanifolds $L_y \subset M$ and disjoint union $\coprod L_y = M$.

To verbalize the notion of foliation is an equivalent notion with integrability by Frobenius theorem, where integrability is done by Lie algebra. If Lie algebra describes the global property (there's no "if" in math Toynbee says), the corresponding Lie group quotient of manifold $[M/G]$ is a stack. It doesn't work, however. I'll use Lie algebroid to describe every point of the submanifold $L \subset M$, whose Lie groupoid $\mathcal{G}$ quotient is a quotient stack, which describes foliation $[M/\mathcal{G}]$. The first step of foliation my plan is to modernize the definition of foliation by using quotient stack with Lie algebroid/groupoid.

Definition 0.2. Groupoid

A groupoid is a category in which every morphisms are invertible. Denote the objects are $Ob(\mathcal{C}) = G_0$, and the morphisms are $Mor(\mathcal{C}) = G_1$, the double arrow $G_1 \rightrightarrows G_0$. For an arrow $g : x \to y$, there are source and target maps $s, t, : G_1 \to G_0$, together with composition $G_1 \times_{G_0} G_1 \to G_1$, identity arrow and inverses. A group becomes a Lie groupoid if G_0 and G_1 are smoooth manifolds, and all the maps are smooth.

There are many important examples of Lie groupoid. The trivials things (logically equivalent to Lie group but somehow interesting generalization) are Lie group, which is a Lie groupoid with one object so $G \rightrightarrows *$. Another example which is more geometric is that Lie group action $G \times X \rightrightarrows X$ can be modeled by $G_1 = G \times X$ and $G_0 = X$. By the context of foliation, Lie groupoid $Hol(\mathcal{F})$ of the foliation $\mathcal{F}$ is holonomy groupoid. All the above examples are important. Now we consider quotient stack by Lie groupoid, in particular by Lie group. The definition of quotient stack below is a moduli stack while keeping the group symmety structure for group G, but groupoid quotient $\mathcal{G}$ is somewhat more complicated:

Definition 0.3. Quotient Stack

For an algebraic variety X, quotient stack by Lie group G is $[X/G]$ is a functor $[X/G] : (Spaces)^{op} \to Grouoids$.

$$[X/G](S) = \{(P \to S, f : P \to X) | P \to S \text{ is a principal G-bundle}, fisG - equivariant\}$$

In particular if $X = pt$, the quotient stack is $[pt/G] = BG$, and $BG(pt)$ is one-object groupoid whose automorphism group is $G = Hom(*, *)$.

Meanwhile, quotient stack by groupoid $[U/R]$ for a groupoid R is a stackification $[U/R] = St([U/_pR])$ of category fibered in groupoids $p : U \to R$. This works for all $c \in R$, there exists $x \in U$ s.t. $p(x) = c$, and for all $f : x \to y$, $p(f) = id_c$.

where I abbreviate what the stackification means in this introductory chapter for the moment. In any case, the important quotient stack is holonomy groupoid quotient $[M/Hol(F)]$, which is a foliation. But how does Lie groupoid appears, where Lie algebra solves integrability condition in manifold? Lie algebra can be generalied to Lie algebroid by its globalization, where Lie groupoid comes. The definition of Lie algebroid is:

Definition 0.4. Lie Algebroid

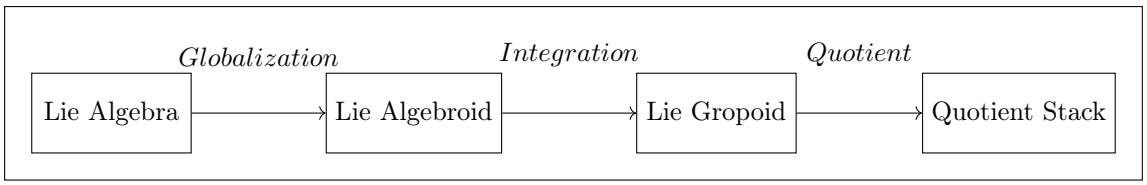
For a smooth manifold M, a Lie algebroid over M consists of $(A \to M, [\cdot,\cdot]_A, \rho)$ where $A \to M$ is a vector bundle, $[\cdot,\cdot]_A : \Gamma(A) \times \Gamma(A) \to \Gamma(A)$ is a Lie bracket of sections, $\rho : A \to TM$ is an anchor map. These satisfy product rule $[X, fY]_A = f[X, Y]_A + \rho(X)(f)Y$, and the anchor preserves bracket $\rho([X, Y]_A) = [\rho(X), \rho(Y)]$.

This Lie algebroid is a generalization of Lie algebra. In fact, the fiber A_x $x \in M$ of vector bundle $A \to M$ is Lie algebra, so containing the data of local symmetry.

Remark 0.1

As a comparison, quotient stack $[M/G]$ of gauge theory is a Lie group quotient, not Lie groupoid, where the local symmetry of particle action is everywhere the same, but the construction is nothing to do with the foliation problem. In other words, not all quotient stack is foliation.
The computation of quotient stack $[M/G]$ is done by Chevalley-Eilenberg complex. There's generalization of Chevalley-Eilenberg for the case of quotient stack by Lie groupoid.

In summary, the above argument can be depicted in the following flow chart, and now that the foliation is modernized geometrically, studying foliation is somewhat equivalent to studying quotient stack:



Several questions appear from this informal writing of foliation. First of all, to make the argument more precisely, can Lie algebroid be always integrated to the Lie groupoid? Now true in general, but Crainic-Fernandes integrability criterion claims Lie algebroid used in foliation can be always integrated.

Proposition 0.1. Crainic-Fernandes Criterion

TBD

We can also discuss foliation using derived algebraic geometrically, so we can discuss resolving singularity problem of foliation and intersection, deformation, and any other algebraic geometric technique.

For defining derived foliation, derived foliation from Toën POV is graded mixed cdga, which means many cdga. The nuance is that set theoretically a foliation is a process of transforming a manifold into a disjoint union of submanifolds, and if a manifold is cdga, then submanifolds are also cdga, and this derived foliation actually corresponds to Lie algebroid.

Definition 0.5. Derived Foliation

Affine derived foliation is graded mixed cdga mapping $Fol : dAff \to Cat_\infty$, and its generalization to any derived stack $X \in dSt$ is projective limit of affine derived foliation $Fol(X) = \varprojlim_{Spec(A)\to\mathcal{F}} Fol(Spec(A))$.

Proposition 0.2. Lie Algebroid Correspondence

There is a canonical mapping

$$LieAlgbd/X \to Fol(X)$$

Once setting is done, moving onto the generalization of derived foliation on singularity, I here list some brief conjectures on Lagrangian, quotient stack, intersection theory, degeneration, and Fukaya category, and these detail are given in each session.

Conjecture 0.1

- This foliaton is complex dimensional, but should be generalized to Lagrangian.
- What's the connection with Lie groupoid quotient stack formulated below?

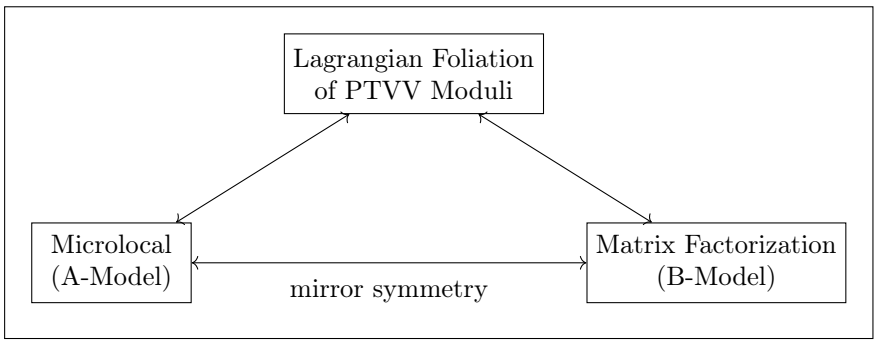
$$LieAlgbd/X \cong [X/\mathcal{G}]$$

- Intersection
  Two quotient projection maps $p : X \to X/\mathcal{F}$ and $q : X \to X/\mathcal{G}$ of foliations, mapping contravariantly $p^* : \mathcal{O}_{X/\mathcal{F}} \to \mathcal{O}_X$ and $q^* : \mathcal{O}_{X/\mathcal{G}} \to \mathcal{O}_X$, which makes a pullback diagram below to make a fiber product $\mathcal{O}_{X/\mathcal{F}\times_X X/\mathcal{G}}$, andn this is the derived intersection:

$$\begin{array}{ccc} \mathcal{O}_{X/\mathcal{F}\times_X X/\mathcal{G}} & \longrightarrow & \mathcal{O}_{X/\mathcal{F}} \\ \downarrow & & \downarrow{\scriptstyle p^*} \\ \mathcal{O}_{X/\mathcal{F}} & \xrightarrow{q^*} & \mathcal{O}_X \end{array}$$

- Degeneration
  If Lagrangian derived foliation contains singularity, its picture might be the same as degeneration around the singular point, whose the parameter moves on quotient stack $[M/\mathcal{G}]$, where usual parameter on degeneration is $\mathbb{R}$ with $t \to 0$.
- Fukaya Category
  Generalizing Nadle-Zaslow $Sh_c(M) \cong Fuk(T^*M)$ to the case of any symplectic geometry, assuming any symplectic geometry has a paraphrase to cotangent bundle or equivalent as $[X/\mathcal{G}] \cong [T^*Y]_{der}$ for some Y.

Finally, I'll leave a conjecture of connecting the previous arguments to the homological mirror symmetry. As noted later, based on Nadler-Zaslow I'll conjecture generalization of Fukaya category of the cotangent bundle to general symplectic geometry either derived intersection problem or sheaf quantization on A-model side, while derived Lagrangian foliation on PTVV moduli space contains cyclic homology of categorified matrix factorization, on B-model side, as shown in the chart below.



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1 Derived Foliation

Definition 1.1. Graded Mixed CDGA

For defining graded mixed cdga, cdga is a graded commutative $xy = (-1)^{|x||y|}yx$ and dg algebra, and let graded mixed object E be a $\mathbb{Z}$-graded object $E = \oplus E(n)$ with extra differential $\epsilon_n : E(n) \to E(n+1)[-1]$ or $\epsilon : E \to E((1))[-1]$ s.t. $\epsilon^2 = 0$. Each $E(n)$ has a differential $d : E(n)^i \to E(n)^{i+1}$. ∞-category of graded mixed objects are denoted by $\epsilon - dg^{gr}$, and if E is also cdga, it forms category $\epsilon - cdga^{gr}$. I'll denote category of derived affine scheme $dAff^{\mathcal{F}} = \epsilon - cdga^{gr}$ as contravariant of that of the graded mixed cdga. In addition, consider an ∞-functor $DR : cdga \to \epsilon - cdga^{gr}$ where $DR(A) = Sym^u_{A'}(\Omega^1_{A'}[1])$. This DR is functorial in the sense of ∞-categories, and analogously, $DR(X) = Sym_{\mathcal{O}_X}(\mathbb{L}_X[-1])$.

Definition 1.2. Derived Scheme

A derived scheme $(X, \mathcal{O}_X)$ consists of functor $\mathcal{O}_X : Open(X)^{op} \to sComm$, mapping $U \mapsto \mathcal{O}_X(U)$ where $sComm$ is an ∞-category of simplicial commutative rings (also called derived rings) defined by $sComm := Fun(\Delta^{op}, CRing)$. This $sComm$ can be realized as a category of graded commutative ring $GrCRing$ mapping $\mathcal{O}_X \mapsto \pi^*(\mathcal{O}_X)$, whose trivialization to the ordinary scheme is $sComm \to CRing$ mapping $\mathcal{O}_X(U) \mapsto \pi_0(\mathcal{O}_X(U))$.

$$\begin{array}{ccc} sComm & \longrightarrow & GrCRing \\ & \searrow & \downarrow \\ & & CRing \end{array} \qquad \begin{array}{ccc} \mathcal{O}_X & \longmapsto & \pi_*(\mathcal{O}_X) \\ & \searrow & \downarrow \\ & & \pi_0(\mathcal{O}_X) \end{array}$$

The truncation of derived scheme t_0 is taking π_0 on $(X, \mathcal{O}_X)$, denoted by $t_0 : dSch \to Sch$, and its inverse is full embedding $i^* : Sch \to dSch$ by using $i : CRing \to sComm$.

$$dSch \underset{i^*}{\overset{t_0}{\rightleftarrows}} Sch$$

One of natural question might be if the scheme $(X, t_0(\mathcal{O}_X))$ can reconstruct the original derived scheme? The existence of higher homotopy by Postnikov tower

$$\mathcal{O}_X \to \cdots \to t_{\leqq n}(\mathcal{O}_X) \to t_{\leqq n-1}(\mathcal{O}_X) \to \cdots \to t_0(\mathcal{O}_X) = \pi_0(\mathcal{O}_X)$$

making a sequence of derived schemes $\{(X, t_{\leqq n}(\mathcal{O}_X))\}$.

We define affine derived foliation using the definition of derive scheme. As noted later, affine derived foliation is a graded mixed cdga, whose category can be identified with derived stack. Intuitively, if foliation is a transformation of a manifold to a set of disjoint union, transformating derived scheme to derived foliation is to transform cdga to graded mixed cdga, which is many cdga.

Definition 1.3. Affine Derived Foliation

An affine derived foliation is a graded mixed cdga A with connectivity, perfectness, and quasi-freeness.

- (Connectivity)
  The underlying cdga $A(0)$ is cohomologically concentrated in non-positive degrees and is finitely presented over $\mathbb{C}$.
- (Perfectness)
  The $A(0)$-dg-module $A(1)[-1]$ is perfect and connective.
- (Quasi-freeness)
  The natural morphism of graded cdga's

  $$Sym_{A(0)}(A(1)) \to A$$

  is a quasi-isomorphism of graded cdga.

Affine derived foliation generates a category $dAff^{\mathcal{F}}$, and it has a canonical ∞-functor $dAff^{\mathcal{F}} \to dAff$ with canonical mapping $A \mapsto Spec(A(0))$.

For the further formalization, an ∞-functor $Fol : dAff^{op} \to Cat_\infty$ desribes all the possible foliations, while $dAff^{\mathcal{F}}$ only describes one of them, namely a foliation is an object $\mathcal{F} \in Fol(X)$. We call this $\mathcal{F}$ as derived foliation.

Definition 1.4. Derived Foliation

$$\begin{array}{ccc} dAff & \xrightarrow{Fol} & Cat_\infty \\ & \searrow & \downarrow \\ & & dAff^{\mathcal{F}} \end{array} \qquad \begin{array}{ccc} Spec(A) & \longmapsto & Fol(Spec(A)) \\ & \searrow & \downarrow \\ & & Spec(A') \end{array}$$

where $A = A'(0)$.

For the generalization of derived affine scheme $Spec(A)$ to derived stack $X \in dSt$, consider scheme is defined to be locally isomorphic to an affine scheme, but categorically reformulated to a projective limit of affine schemes.

Accordingly, the foliation will be rewitten as

$$Fol(X) = \varprojlim_{Spec(A)\to\mathcal{F}} Fol(Spec(A))$$

where for definining derived stack dSt, we'll start from classical stack, and roughly speaking, stack is a 2-sheaf, in other words a functor from category of vector bundles to some category with Grothendieck topology and descent. Typically the codomain would be ∞-groupoid. For derived argument, derived prestack is category $P(dAff_k^{op}) = Fun(dAff_k^{op}, S)$, and derived stack is an embedding $Shv(dAff_k, \tau) \hookrightarrow Fun(dAff_k^{op}, S)$, this embedding functor has an adjoint, called stackification. This topology τ defines an accessible left-exact localization, this is a presentable ∞-category, uses as a replacement of Grothendieck topology.

Definition 1.5. Derived Stack

TBD

Introduce example of derived foliation.

Definition 1.6. Lie Algebroid

For a smooth DM stack X, its tangent sheaf $\mathbb{T}_X$ is a sheaf of Lie algebras on the small étale site X_{et}, where the Lie bracket is the usual bracket of vector fields. Lie algebroid consits of a vector bundle $T \to X$ together with

- A $\mathbb{C}$-linear Lie bracket $[-,-]$ on T.
- an $\mathcal{O}_X$ linear morphism $a : T \to \mathbb{T}_X$
- $[s, ft] = f[s,t] + a(s)(f)t$

One of the applications of Lie algebroid is that category of Lie algebroid over X corresponds to $Fol(X)$

$$LieAlgbd/X \to Fol(X)$$

Consider Lie algebra associative correspondence $\wedge V[1] \cong Sym(V)[1]$, Lie algebroid similarly has a correspondence $C^*(T) = Sym_{\mathcal{O}_X}(T^*[1])$ By noticing that the bracket induces the derivation just a mapping $d : T^* \to T^* \wedge_{\mathcal{O}_X} T^*$, the complex $C^*(T)$ is idenfied as cdga. We called this $C^*(T)$ as Chevalley-Eilenberg complex.

Question 1.1

- Derived foliation of formal scheme is still formal? Yes.
- Is there any extension of foliation to Lagrangian foliation? To answer this question, all the previous foliation is complex dimension.

Definition 1.7. Perfect Crystal

Crystal is a vector bundle with flat connection. Perfect crystal is a crystal whose underlying complex is perfect.

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2 Cotangentification

I'll consider extending derived foliation to Lagrangian foliation, and I'll begin with Symplectic vector space and its generalization to derived algebraic geometry to build some intuition. Consider(informal) Lagrangian quotient in place of Lie groupoid quotient. Let Symplectic vector space be V with its 2-form ω and some Lagrangian $L \subset V$ s.t. $\omega|_L = 0$, the quotient V/L as a vector space is the baby form of Lagrangian foliation problem, where we have further generalization:

Definition 2.1

- step 1 (vector space):
  The quotient V/L is no longer a symplectic space, but it induces the pair $L \times V/L \to \mathbb{R}$ in addition the duality $L \cong (V/L)^*$ make it $(V/L)^* \times V/L \to \mathbb{R}$.
- step 2 (cotangent bundle):
  For a smooth manifold X, cotangent bundle T^*X is a symplectic manifold.
- step 3 (symplectic manifold):
  For the underived case, let a symplectic manifold X and its tangent bundle TX and Lagrangian L, the quotient TX/L replaces (V/L).
- step 4 (derived method):
  Globally, we use L_∞ groupoid quotient stack $[X/\mathcal{G}]$, and derived tangent complex $\mathbb{T}_{[X/\mathcal{G}]} \cong [\mathfrak{g}f \to TX]$ or noted by $\mathbb{T}_{\mathcal{F}}$ of derived foliation. the duality pairing will be $\mathbb{T}_{\mathcal{F}}^{\vee} \otimes \mathbb{T}_{\mathcal{F}} \to \mathcal{O}L_X$.

For the last part, consider the two distinct Lagrangian foliation $\mathcal{F}$ and $\mathcal{G}$ on symplectic space M, and assuming each leaves of two foliation are transversal. Then the tangent bundle will be a direct sum $TM = T\mathcal{F} \oplus T\mathcal{G}$. If foliation creates quotient stack, is there any duality to $[M/G]$, namely, $T[M/G] \cong T\mathcal{G}^*$?

Conjecture 2.1

$$T[M/G] \cong T\mathcal{G}^*$$

Question 2.1

Is there any geometrc realization of $Perf(\mathcal{F})$?

So far we defined Symplectic calculus using Lagrangian quotient, which works fine on vector space, but it doesn't work in general, since not all leaves on the Lagrangian foliation are the same. But is it still possible to trivialize Lagrangian foliation, so each leaves of that foliation is isomorphic to one fixed foliation up to deformation or any other? Locally yes, but it doesn't work in general for obstruction of holonomy and topology. However, supposely we assume very naïvely an identification of the quotient stack of foliation and some derived cotangent analogue in this formula [1], cotangent bundle has particularly important of general symplectic manifold. Below is a conjecture:

Conjecture 2.2. Symplectic Cotangent

$$[X/\mathcal{G}] \cong [T^*Y]_{der}$$

[1] prove it

Now to introduce Nadler-Zaslow as an application of cotangent bundle T^*X, which is an analogue of HMS for the case of $Fuk(T^*X)$ using microlocal analysis.

Proposition 2.1. Nadler-Zaslow

There is an equivalence of category

$$Fuk(T^*X) \cong Sh_c(X)$$

where c means constructible not compactness.

Consider Fukaya category doesn't need to be symplectic manifold, but could also be symplectic stack, and consider its generalization to n-shifted cotangent bundle of moduli space $T^*[n]\mathcal{M} = Spec_{\mathcal{M}}Sym(\mathbb{T}_{\mathcal{M}}[-n])$.

Conjecture 2.3. Nadler-Zaslow Enhanced

$$Fuk(T^*[n]\mathcal{M}) \cong Sh_c(\mathcal{M})$$

Remark 2.1

Consider the generalization of microlocal support to moduli stack. The microlocal support $SS(\mathcal{F}) \subset T^*[n]\mathcal{M}$, as a Lagrangian submanifold, and mapped from a sheaf $\mathcal{F}$.

It'll be also interesting to consider extracting Symplectic structure from moduli stack by some projection $p : T^*[n]\mathcal{M} \twoheadrightarrow X$, but the problem is Fuk cannot be an ∞-functor $Fuk(p) : Fuk(T^*[n]\mathcal{M}) \twoheadrightarrow Fuk(X)$.

Conjecture 2.4

Discuss functoriality of Fuk.

Consider Tamarkin category and sheaf quantization.

3 Degeneration

Degeneration in algebraic geometry is and operation to take limit in geometric object. The famous example is pinched torus is degeneration of torus family by taking limit, which is in other words, adding neighborhood torus family on the pinched torus, to understand more structure on itself. Another important case is reducible degeneration, where the degenerated geometry can be decomposed to the multiple objects. An important example is moduli space of Rieman surface $\mathcal{M}_{g_1+g_2,n_1+n_2-1}$ is decomposed to the union of $\mathcal{M}_{g_1,n_1}$ and $\mathcal{M}_{g_2,n_2}$ sharing a point, which we call it nodal degeneration. Another example of higer dimensional case is Tyurin degeneration separating K3 suface X to the union $X \rightsquigarrow X_1 \cup_S X_2$, where the two objects intersect $X_1 \cap X_2 = S$, where S is a torus.

Example 3.1. Illustration

Put picture here.

This time, my goal here is to define degeneration of Lagrangian foliation $\mathcal{F}$, assuming existence of singularity both in its moduli space and itself, and how to parametrize degeneration on PTVV moduli space of Fano manifold X or on the quotient stack $[M/\mathcal{G}]$.

Definition 3.1. Degeneration

A degeneration is a mapping

$$f : \mathbb{A}^1 \to \mathcal{M}$$

where $\mathbb{A}^1$ is an affine plane and fibers $\pi^{-1}(t)$ form a family of varieties over $\mathbb{A}^1$. $\pi^{-1}(0)$ can be thought as the limit. More generally, the affine plane $\mathbb{A}^1$ can be replaced to other algebraic varieties.

Definition 3.2. PTVV Moduli Stack

TBD.

Definition 3.3. Quotient Stack

TBA quotient stack $[\mathcal{M}/\mathcal{G}]$? it contains holonomy and isotropy. is there any singular holonomy?

Tyurin degeneration and matrix factorization. PTVV moduli space of Fano manifold.

Remark 3.1

As an assumption, Fano manifold argument can be generalized to any Kähler manifold. At least, Fano manifold can be transformed to log Calabi-Yau by deformation theory.

Definition 3.4

4 Derived Intersection

One motivation of derived intersection is to extract some symplectic geometry from PTVV moduli space through derived Lagrangian intersection.

Two quotient projection maps $p : X \to X/\mathcal{F}$ and $q : X \to X/\mathcal{G}$ of foliations, mapping contravariantly $p^* : \mathcal{O}_{X/\mathcal{F}} \to \mathcal{O}_X$ and $q^* : \mathcal{O}_{X/\mathcal{G}} \to \mathcal{O}_X$, which makes a pullback diagram below to make a fiber product $\mathcal{O}_{X/\mathcal{F}\times_X X/\mathcal{G}}$, andn this is the derived intersection:

$$\begin{array}{ccc} \mathcal{O}_{X/\mathcal{F}\times_X X/\mathcal{G}} & \longrightarrow & \mathcal{O}_{X/\mathcal{F}} \\ \downarrow & & \downarrow{\scriptstyle p^*} \\ \mathcal{O}_{X/\mathcal{F}} & \xrightarrow{q^*} & \mathcal{O}_X \end{array}$$

deformation why deform lie groupoid not stack?

group action

Definition 4.1

Intersection theory between quotient stacks.
Consider grouopid action functorially. $Fun(BG, X)$
Quotient stack has a

Example 4.1. Equivariant Intersection – Group

Consider a mapping $X \to [X/G]$ where G is a group.

Example 4.2. Equivariant Intersection – Groupoid

Definition 4.2

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